



FAIR-Ready AI

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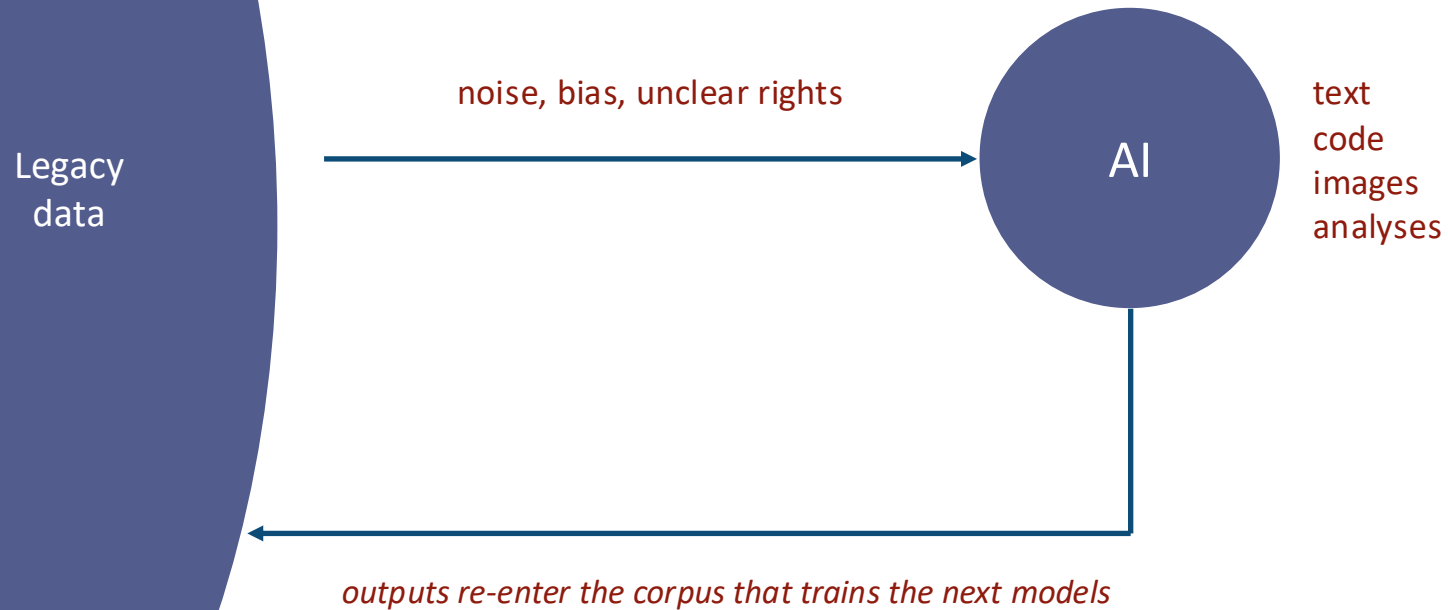
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Anonymous bytes in, anonymous bytes out



What is usually missing

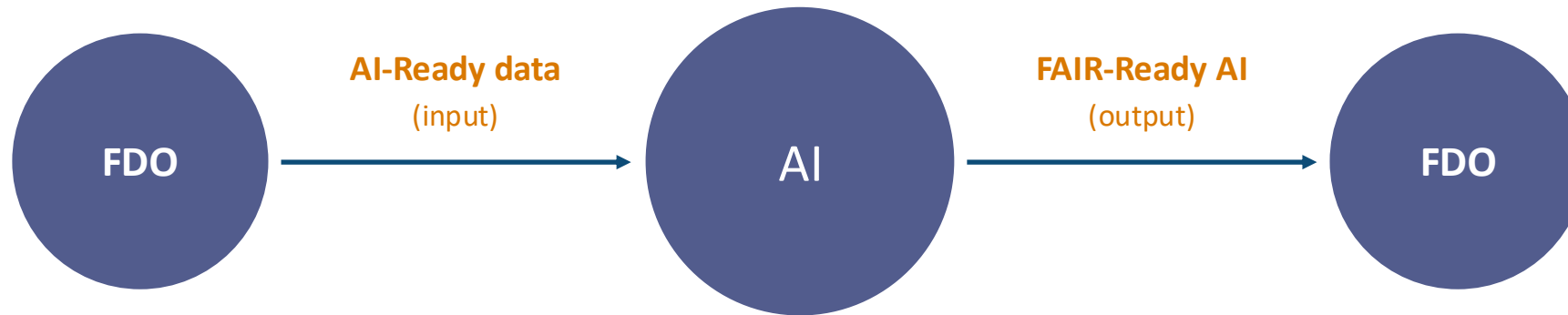
- A persistent identifier
- Provenance: who or what made it
- A licence
- A checksum that fixes the bytes

Recursive training on generated data degrades models: Shumailov et al., Nature 631, 755–759 (2024).
EU AI Act, Art. 50(2): outputs of generative AI systems to be marked as AI-generated in a machine-readable format.

Schultes, E. (2025, October 3). FAIR and verifiable: The role of FAIR Digital Objects in trusted AI. Zenodo. Open Science Conference 2025, Hamburg. <https://doi.org/10.5281/zenodo.17257168>

FDOs in, FDOs out

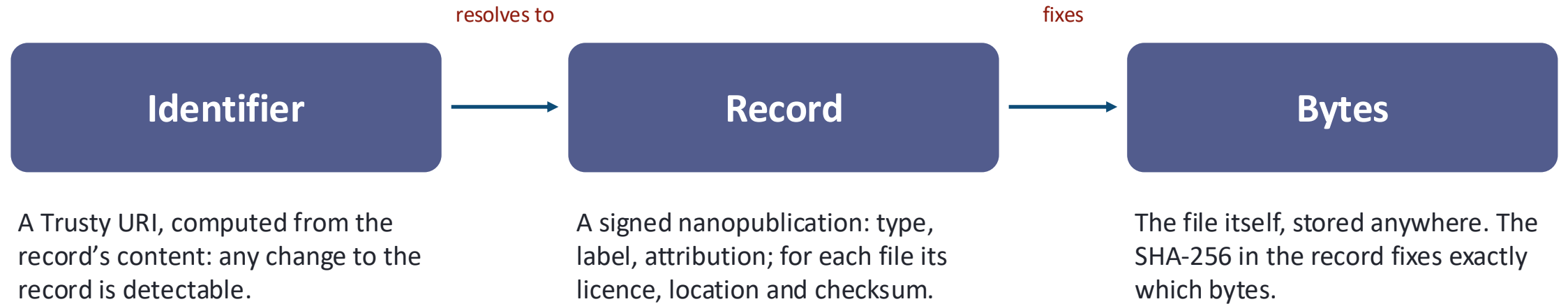
In FAIR AI, the only inputs are FAIR Digital Objects and all outputs are FAIR Digital Objects.



FAIR Digital Object (FDO): a persistent identifier that resolves to a machine-readable record stating what the object is, exactly which bytes it is, who made it, and how it may be reused.

Goal (argued for, not yet a result): minting an FDO becomes as routine as saving a file; one day, a right-click on the desktop.

Nanopublication FDOs: identifier, record, bytes



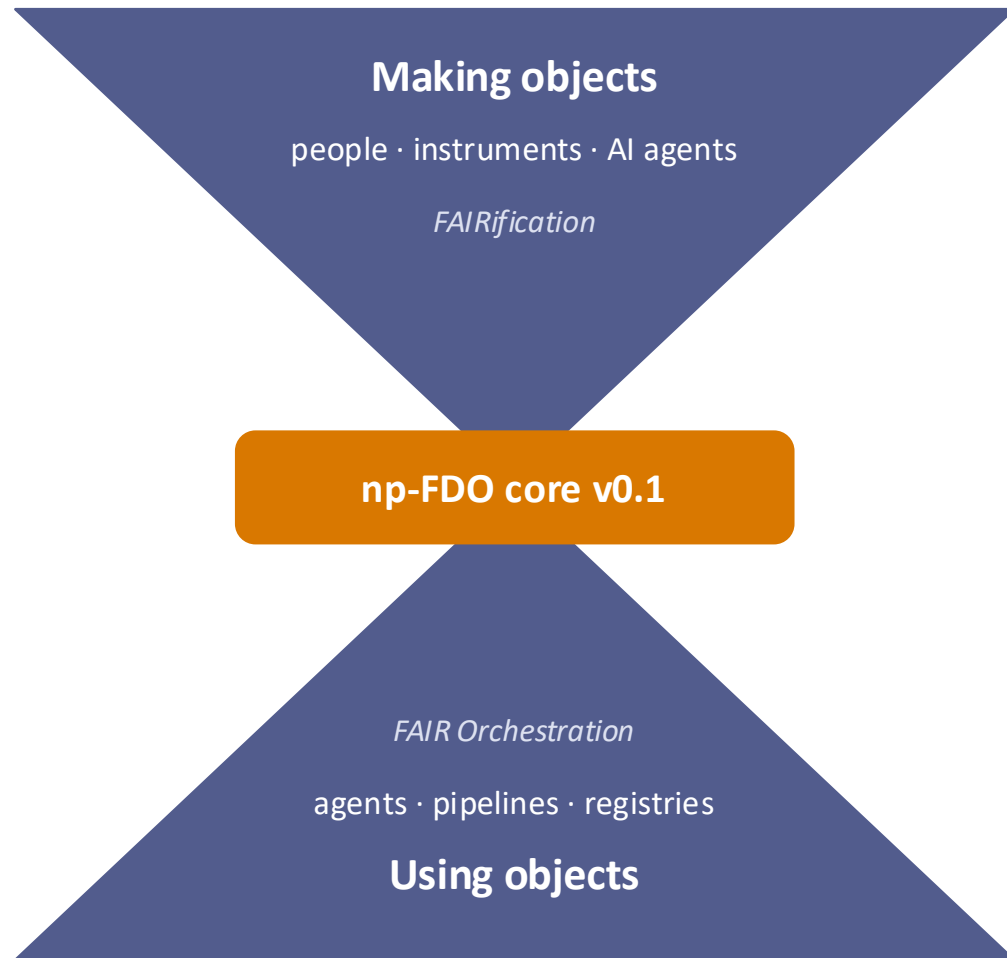
Example: np-FDO core specification v0.1

identifier `https://w3id.org/np/RAYTT7oW0dbPdx3PqIA1m9XBLRGYVqmAhzLY7vLTMG1GI`

bytes `SHA-256 fde7fc20...a8c140 · github.com/easchultes/npfdo-core, commit b400634 · CC BY 4.0`

A machine can verify that the bytes it received are the bytes the record describes.

Nanopublication FDOs: hourglass architecture



Carrier: nanopublications, served by the nanopublication server network

Plus six unconditional FDO constraints: C1 type · C2 label · C3 attribution · C4 marker · C5 identity link · C6 conforms to

Plus one per materialisation: M1 conditions of use (licence)

- Reuse first: no new predicates; three classes minted
- One generic template; the object type is a controlled value
- Designed to be carrier-neutral; implemented first on nanopublications
- Open, CC BY 4.0.
- Not yet: validation shapes

AI appears twice: making objects at the top, using them at the bottom. Both pass through the same waist.

Schultes, E. (2023). The FAIR hourglass: A framework for FAIR implementation. FAIR Connect: Empowering Data Stewardship, 1(1), 13–17. <https://doi.org/10.3233/FC-221514>

What becomes an FDO: three kinds of object

Epistemic

What is the case?

Tested against evidence; can be false

- Dataset (an LC-MS run)
- Assay or analysis result
- A scientific claim
- Slide deck (type minted at v0.1)

Deontic

What is permitted?

Tested against authority and procedure;
can be invalid, not false

- Licence
- Access or use policy
- Informed-consent terms
- Data-use agreement

Agentic

What can be done?

Tested by execution

- Skill (type minted at v0.1)
- Analysis workflow
- Standard operating procedure
- Service or endpoint

The record itself stays epistemic: it states facts about the object and reaches permissions and procedures only by reference to typed objects. Resolving the type tells an agent which kind of statement it is handling.

Specification design decision D20. The designation of types by kind is a proposal for the type catalogue, not part of core v0.1.

Implementation: the v0.1 corpus on the public network

8

records published:

- the specification
- two templates
- an example
- three vocabulary terms
- this symposium

4

checks before / after publication:

- hash gate
- expected Trusty URI
- empty issue list
- fetch-back and re-verification

2

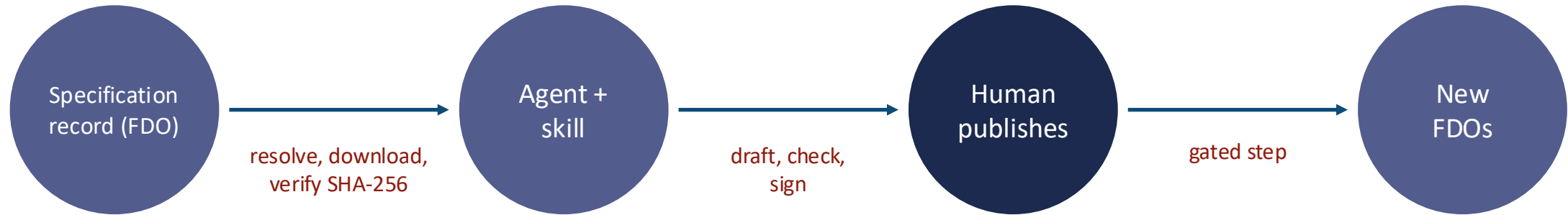
errors caught before publication, by
checks run outside the agent

- The specification is itself an FDO and conforms to itself.
- The vocabulary sits inside the layer it defines: each term is its own record.

Human-controlled publication is necessary, not decorative.

Public minting log: github.com/easchultes/npfdo-core, MINTING_LOG.md

Agent Skills for FAIR AI



Experiment: a fresh agent session with no build context, given the skill, its declared dependency, the network and a task prompt, was asked to create FDOs.

Outputs: vocabulary terms; the skill's own record; a draft of this slide deck.

Observed	Five live records, signed with the corpus key. No published record needed correction.
Skill alone (minimal request)	A conformant core record, live, from a one-line request; type-specific fields still need the prompt.
Corrections	Eleven findings (one major, five moderate, five minor); remedies scheduled for v0.1.1 and v0.2.

Sources: acceptance ledger, npfdo Chat D, and skill-alone test, 28 September 2026. Skill-alone record: w3id.org/np/RA1UII3Z7SzOE0H1RJGw3jY9rlvdMKjYfLI815pArjCsU

Declaring AI involvement



FAIR AI Attribution (FAIA) declares AI involvement in machine-readable form: a flag, an activity code, and the AI system with its version. Relevant to EU AI Act Art. 50(2).

Nanopub Graph	About	Records how...	Activity code
Assertion	the object (the term; the skill; the deck)	the thing itself was made 	Co-Creation
Provenance	the minting activity	its description was produced 	Contribution
Publication information	the record	the container was made 	Contribution

One record separates the AI’s role in the work from its role in the metadata.

- In five live records (28 September); system Claude (claude-opus-5-5); FAIA vocabulary 0.7.0. Values declared by the author, not machine-derived.
- Placement is a proposal to the FAIA community; it enters the core at v0.2.

FAIA: faia.liccium.com; terms reused at w3id.org/faia-framework. Provenance and publication-information statements verified by reading; the conformance checker covers the assertion graph only. Le Dévédec, S. E., Hettne, K., Gambardella, A., Schultes, E., & Posth, S. (2026). FAIA - FAIR AI Attribution White Paper (Version 0.9). Zenodo. <https://doi.org/10.5281/zenodo.20083769> FAIA logo and badges: © FAIA authors, used with permission.

AI-assisted FAIRification

Chat assistant

conversational, skill uploaded

The AI

Drafts and checks the record from a request

Human control

The human signs locally, afterwards: the private key never goes to a remote sandbox

Not yet tested

Local coding agent

terminal

The AI

Drafts, checks and signs with the human's local key; stops at a signed, verified file

Human control

Publication is a separate, human-run, gated step

Used in the acceptance test

Desktop app

local signing service

The AI

Drafts from a conversational request or a right-click

Human control

The human confirms in a dialogue; the key stays on the machine

Designed, not built

The skill makes it easy; the signature, always with the human's key, makes someone accountable.

Where it stands

Demonstrated

Outputs as FDOs: the published corpus, and records minted by an AI agent, each carrying a declaration of AI involvement (FAIA) in all three graphs. An input as an FDO: the agent started from the specification record and verified the bytes against its checksum before use.

Designed, not built

FDO resolution as the standard input channel for agents (MCP endpoint). Desktop minting through a local signing service.

Goal, argued for

FAIR Digital Objects as the accepted standard for the inputs and outputs of FAIR AI.

Next: type-specific fields from the skill alone (slot list shipped with the skill, v0.2) · FAIA placement with the FAIA community · a catalogue of object types · more skills · the MCP endpoint · desktop minting

An invitation: a free, open skill library for FAIR AI

The first skill: npfdo-mint v0.1.0

Makes an AI agent's outputs FDOs: typed, licensed, attributed, signed, checksummed, resolvable. Code Apache 2.0; documents CC BY 4.0.

The skill is itself an FDO

w3id.org/np/RAC05VQyaDSUgzTuGmgdUTh6lklXKBMYo6rnPuxadRZUI

Wanted

Skills for other object types, carriers and agents; types for your own objects; critical review of the core; a view on where AI attribution belongs.



npfdo-skills

github.com/easchultes/npfdo-skills

github.com/easchultes/npfdo-core

In FAIR AI, the only inputs are FDOs and all outputs are FDOs.

All records described here

#1 Specification record: np-FDO core v0.1

<https://w3id.org/np/RAYTT7oW0dbPdx3PqIA1m9XBLRGYVqmAhzLY7vLTMG1GI>

#2 Generic assertion template

<https://w3id.org/np/RA5OrG1-PCAA3Z4kTLKTxq8lhgNAsopAtQM5h0-xj3MSA>

#3 Publication-information template

https://w3id.org/np/RAQV2LjRzG8BQ8wXb6Kbyytc8vVAUeS4pFdoOkG2WA_yI

#4 Example record

https://w3id.org/np/RAE8Y_jz4pP3qZrKg1_zdkD-KvuVraVyfgpTrmaCgnIFQ

#5 Term record: FDORecord (v2, with FAIA)

https://w3id.org/np/RAuMVooxAx5HOD1j7jCjP1RyA_I-spO5f9hdHoQxXFrJQ

#6 Term record: SlideDeck (v2, with FAIA)

https://w3id.org/np/RAMYZAjX-E2HEj_3a9OgssY_14xotscCfs76Ee1TedVmI

#7 Term record: Skill (v2, with FAIA)

<https://w3id.org/np/RATvgMiFNNW1FHeWIDQNS8Pj2rfU95o592p4Ywtic15g>

#8 Conference record: this symposium

<https://w3id.org/np/RAwBRuyJaw0ogGwUTx18woy3nHPcFvDGCXBUXSQJfhZVc>

#10 Skill record: npfdo-mint v0.1.0

<https://w3id.org/np/RAC05VQyaDSUgzTuGmgdUTH6lklXKBMYo6rnPuxadRZUI>

#11 Slide deck record: draft v0.2 of this deck (skill-alone test)

<https://w3id.org/np/RA1U113Z7SzoE0H1RJGw3jY9rlvdMKjYfLl815pArjCsU>